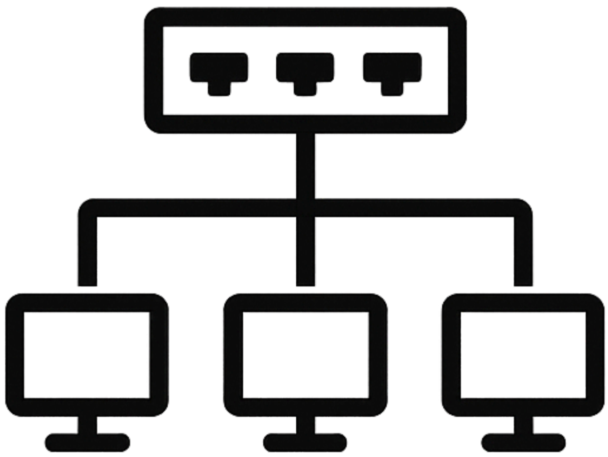
Colombe LANKAR
SIO1

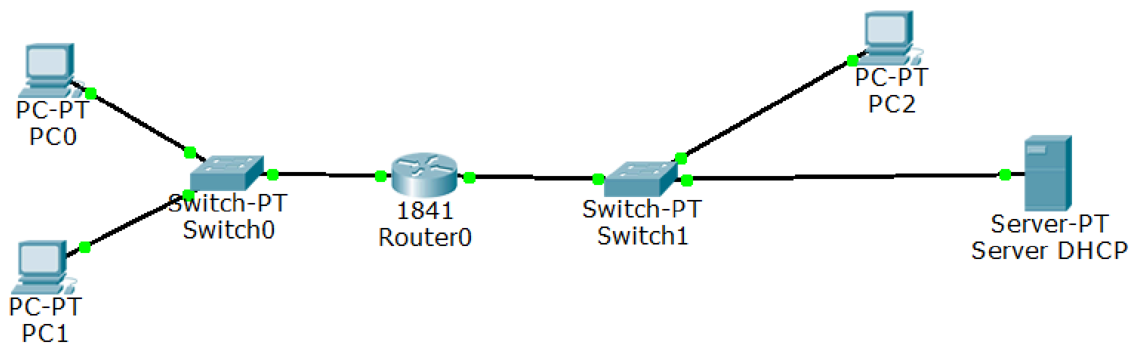


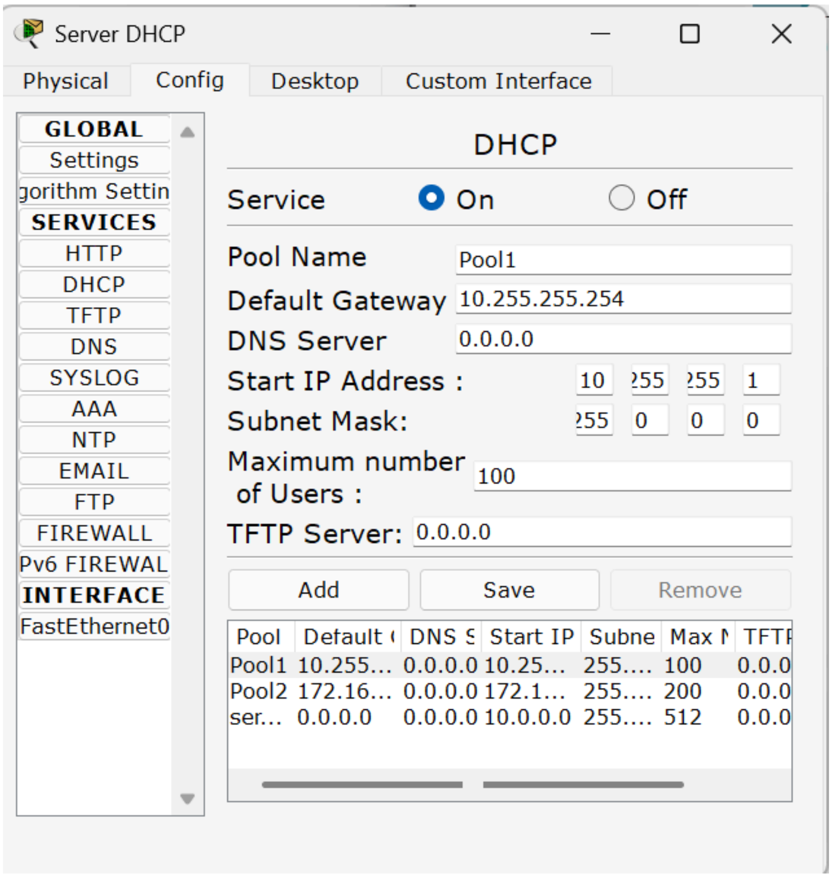
TP RÉSEAU

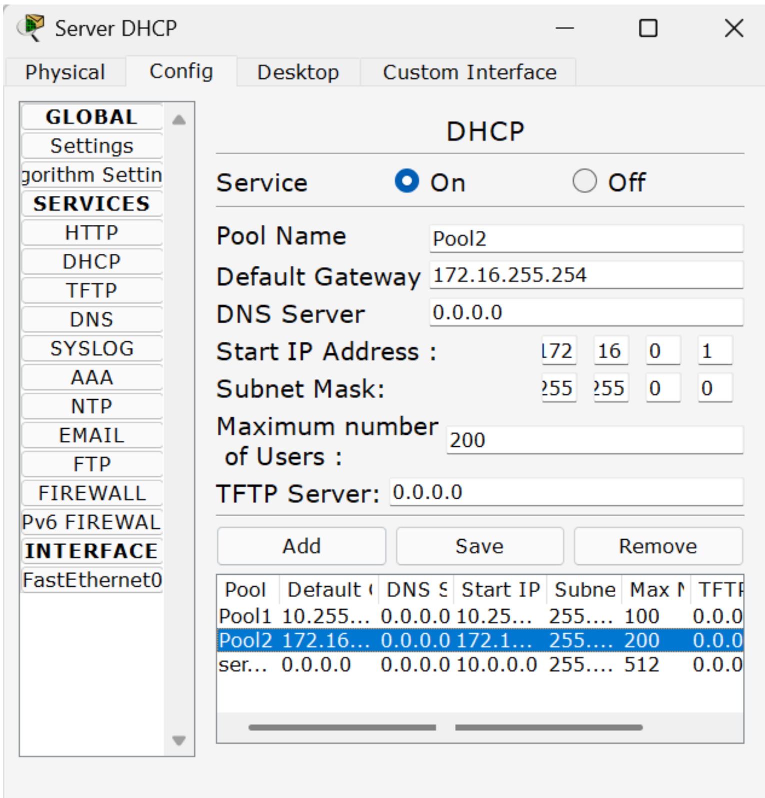
<u>Objectifs :</u>

Séparer le réseau en VLAN10 et VLAN20 et distribuer de IP avec un serveur DHCP

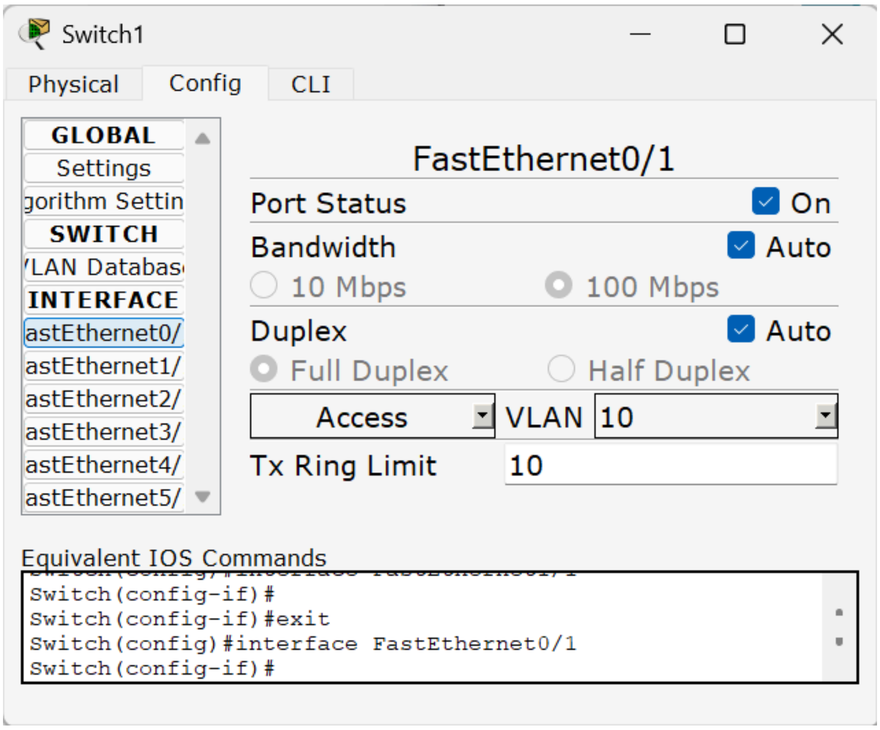
<u>Topologie :</u>

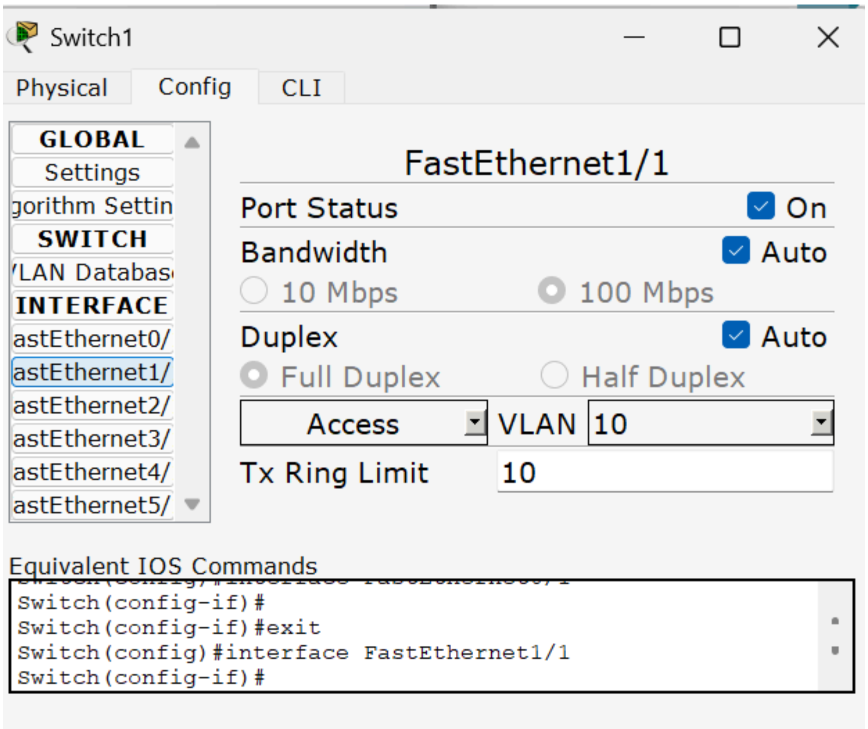


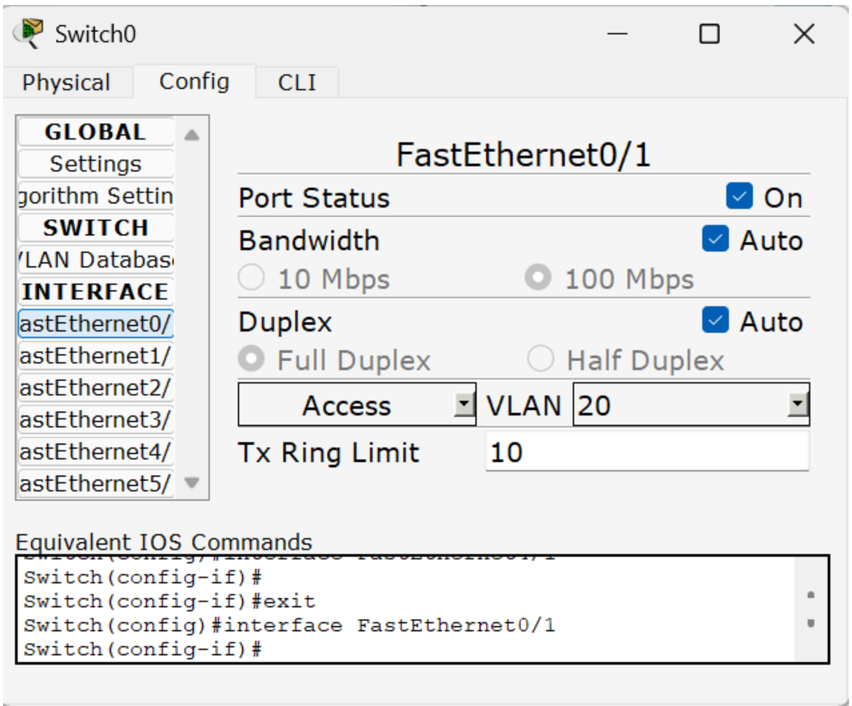


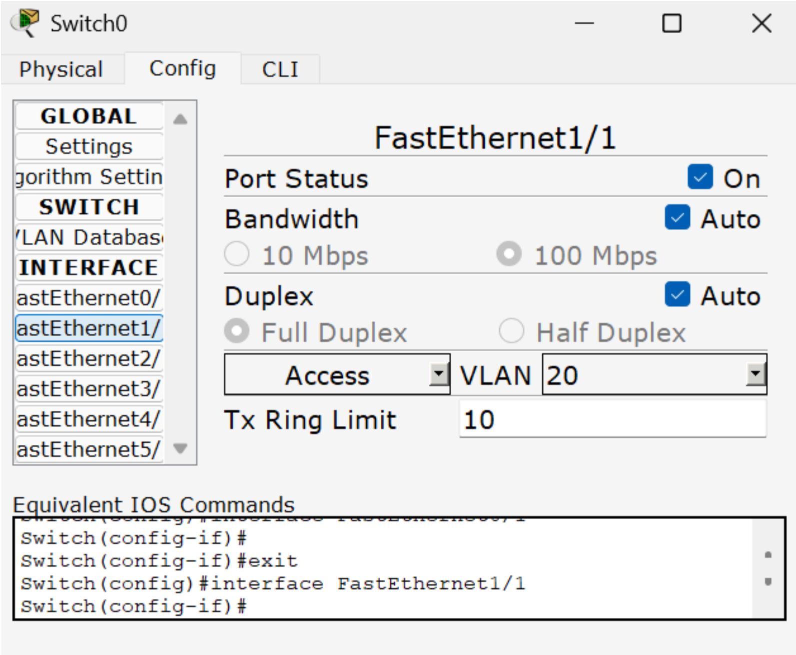


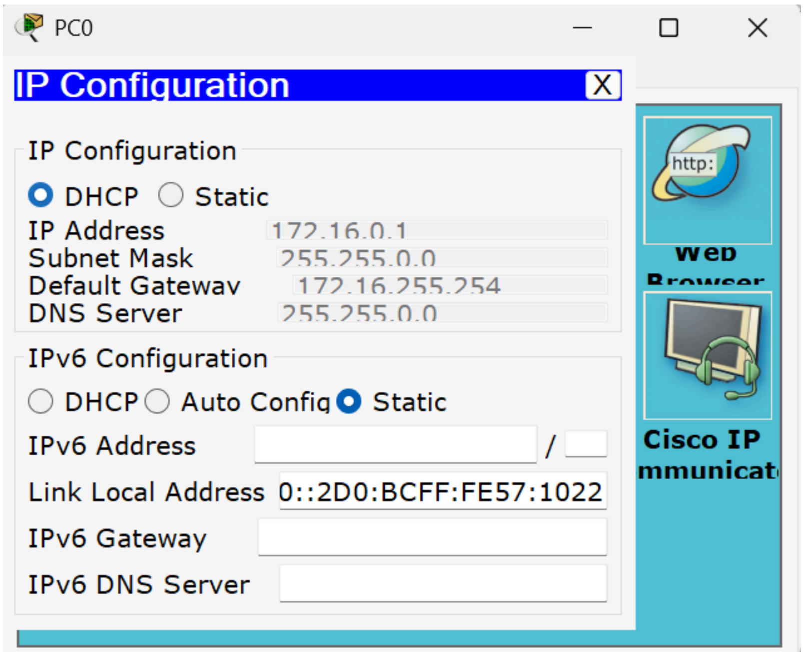
Le serveur DHCP est activé et a deux pools diférents.

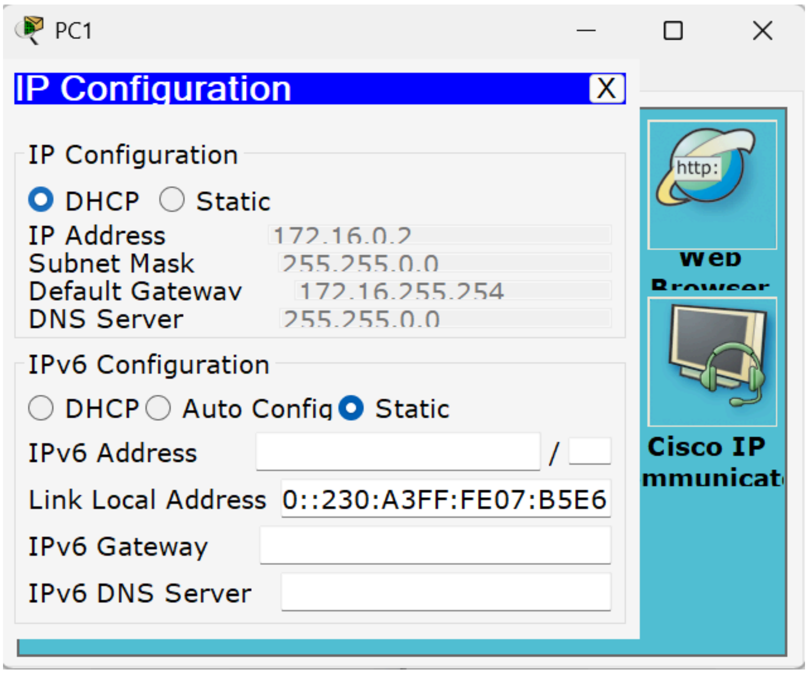


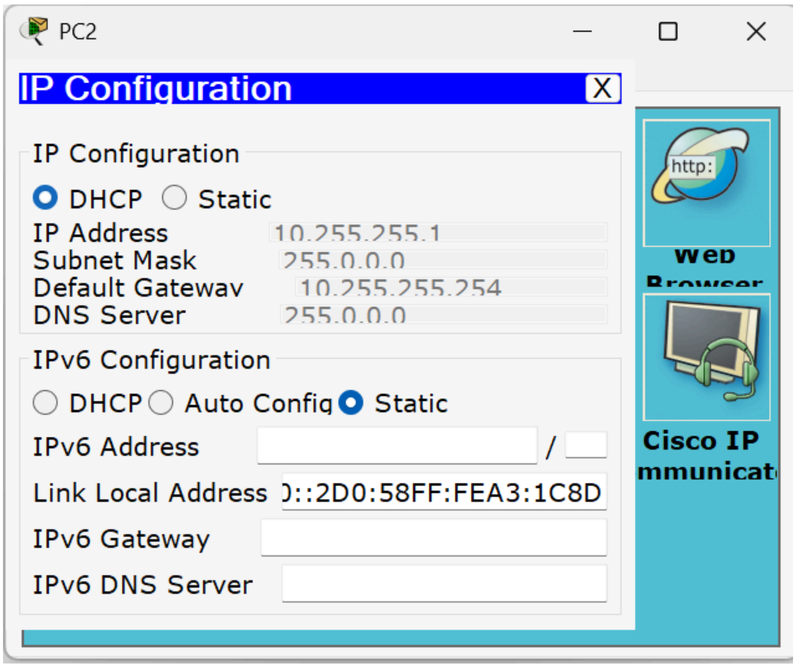

Switch1
Physical
Config
CLI
GLOBAL
Settings
gorithm Settin
SWITCH
'LAN Databas
INTERFACE
astEthernet0/
astEthernet1/
astEthernet2/
astEthernet3/
astEthernet4/
astEthernet5/
FastEthernet1/1
Port Status
On
Bandwidth
Auto
10 Mbps
100 Mbps
Duplex
Auto
Full Duplex
Half Duplex
Access
VLAN
10
Tx Ring Limit
10
Equivalent IOS Commands
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet1/1
Switch(config-if)#

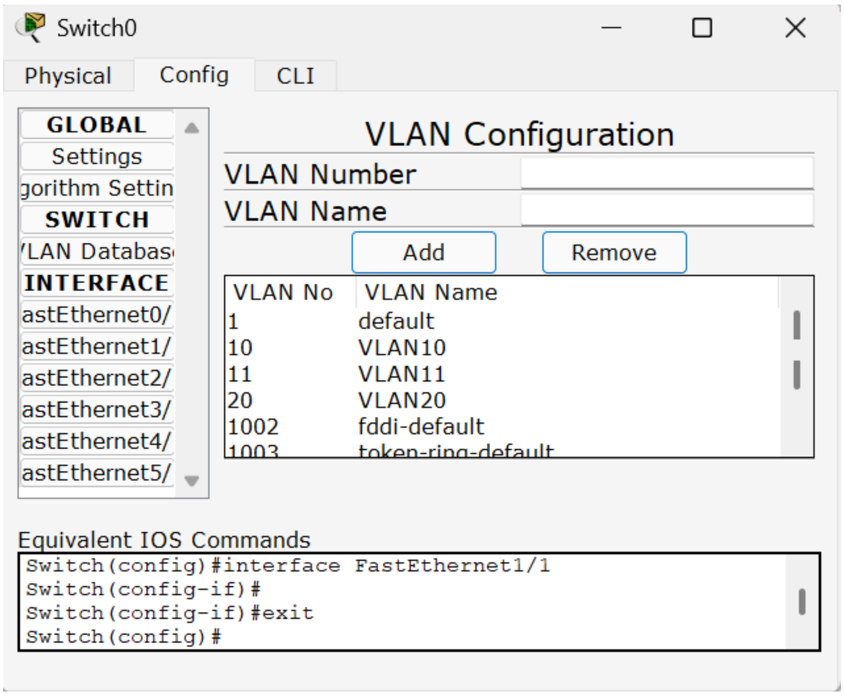

Switch0
Physical
Config
CLI
GLOBAL
Settings
gorithm Settin
SWITCH
'LAN Databas
INTERFACE
astEthernet0/
astEthernet1/
astEthernet2/
astEthernet3/
astEthernet4/
astEthernet5/
FastEthernet0/1
Port Status
On
Bandwidth
Auto
10 Mbps
100 Mbps
Duplex
Auto
Full Duplex
Half Duplex
Access
VLAN
20
Tx Ring Limit
10
Equivalent IOS Commands
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#

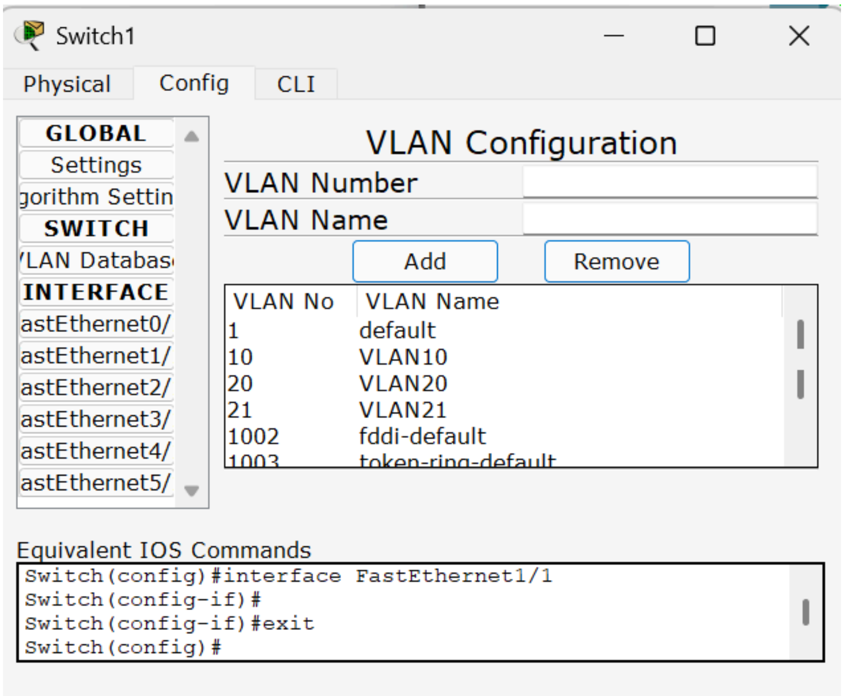

Switch0
Physical
Config
CLI
GLOBAL
Settings
gorithm Settin
SWITCH
'LAN Databas
INTERFACE
astEthernet0/
astEthernet1/
astEthernet2/
astEthernet3/
astEthernet4/
astEthernet5/
FastEthernet1/1
Port Status
On
Bandwidth
Auto
10 Mbps
100 Mbps
Duplex
Auto
Full Duplex
Half Duplex
Access
VLAN
20
Tx Ring Limit
10
Equivalent IOS Commands
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet1/1
Switch(config-if)#

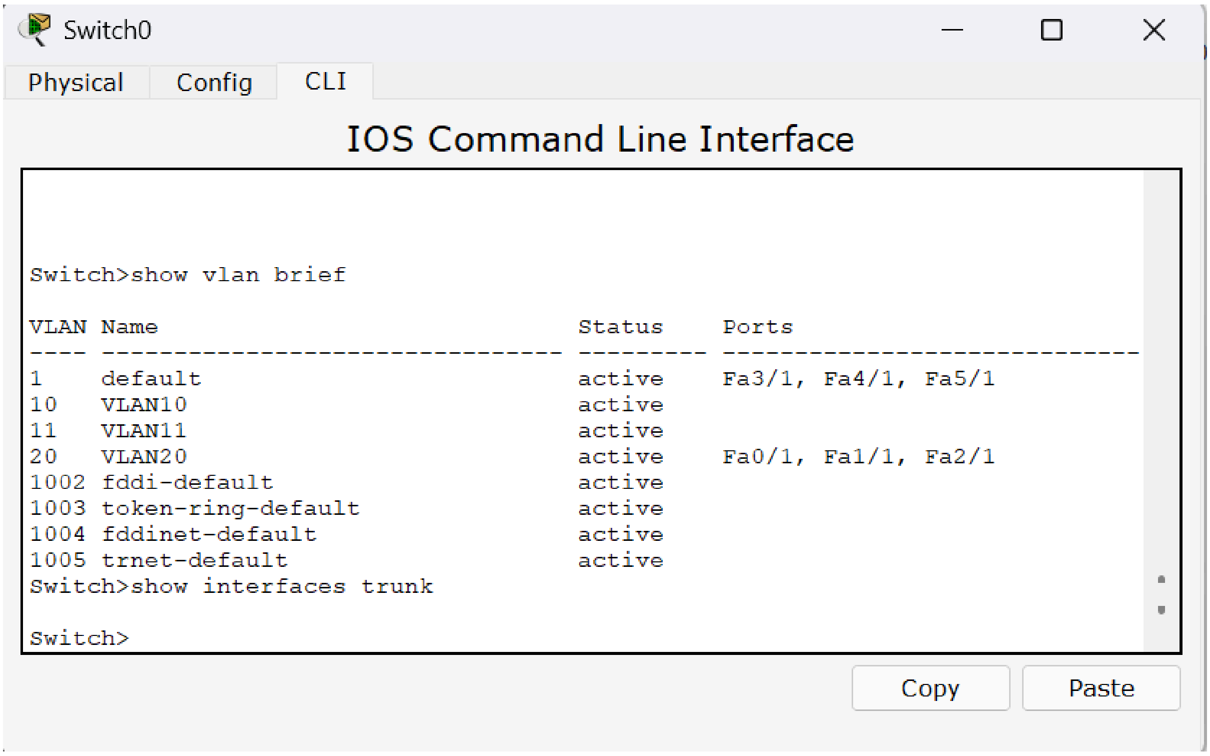

PC0
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
172.16.0.1
Subnet Mask
255.255.0.0
Default Gateway
172.16.255.254
DNS Server
255.255.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
0::2D0:BCFF:FE57:1022
IPv6 Gateway
IPv6 DNS Server
http:
Web
Browser
Cisco IP
mmunicat

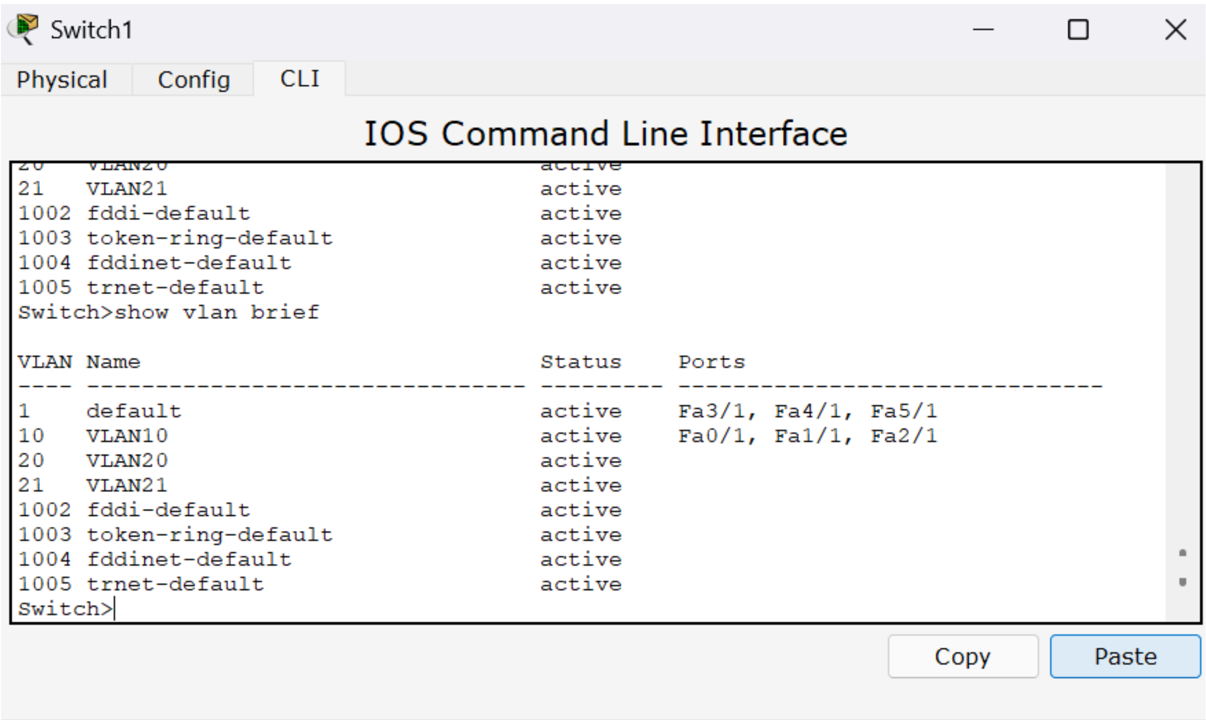

PC1
IP Configuration
IP Configuration
DHCP
Static
IP Address
172.16.0.2
Subnet Mask
255.255.0.0
Default Gateway
172.16.255.254
DNS Server
255.255.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
0::230:A3FF:FE07:B5E6
IPv6 Gateway
IPv6 DNS Server
http:
Web Browser
Cisco IP
mmunicat


PC2
IP Configuration
IP Configuration
DHCP
Static
IP Address
10.255.255.1
Subnet Mask
255.0.0.0
Default Gateway
10.255.255.254
DNS Server
255.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
0::2D0:58FF:FEA3:1C8D
IPv6 Gateway
IPv6 DNS Server
http:
Web Browser
Cisco IP
mmunicat

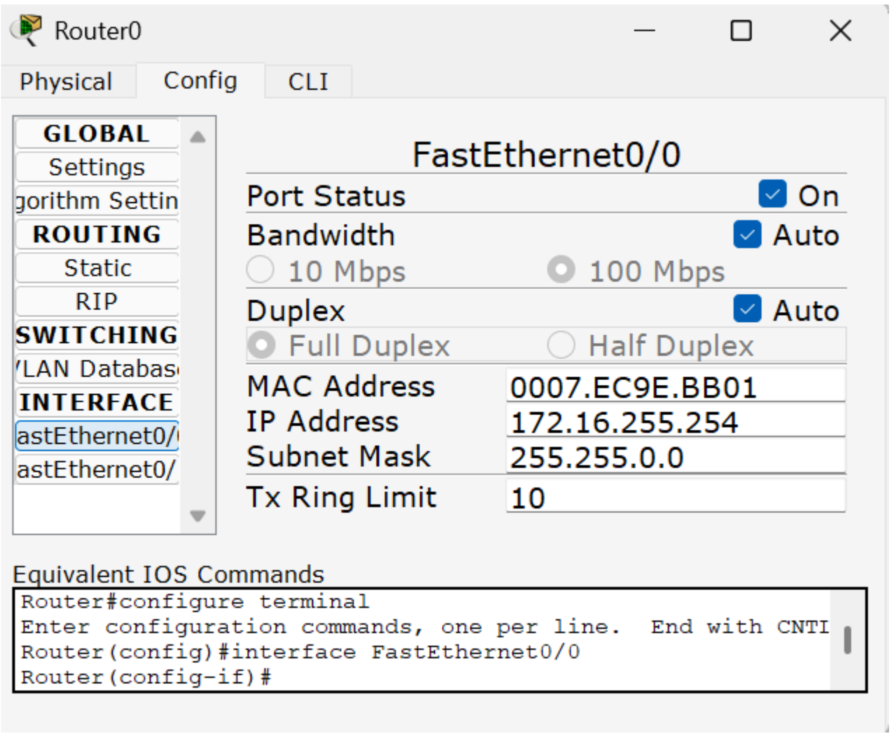

Switch0
Physical
Config
CLI
GLOBAL
Settings
gorithm Settin
SWITCH
'LAN Databas
INTERFACE
astEthernet0/
astEthernet1/
astEthernet2/
astEthernet3/
astEthernet4/
astEthernet5/
VLAN Configuration
VLAN Number
VLAN Name
Add
Remove
VLAN No VLAN Name
1 default
10 VLAN10
11 VLAN11
20 VLAN20
1002 fddi-default
Equivalent IOS Commands
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#

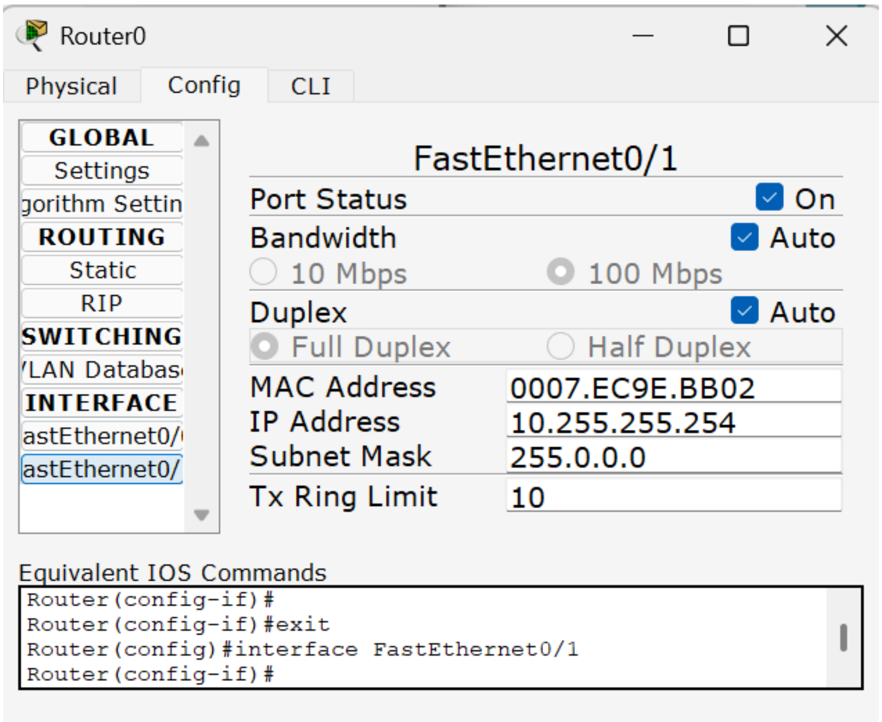

Switch1
Physical
Config
CLI
GLOBAL
Settings
gorithm Settin
SWITCH
'LAN Databas
INTERFACE
astEthernet0/
astEthernet1/
astEthernet2/
astEthernet3/
astEthernet4/
astEthernet5/
VLAN Configuration
VLAN Number
VLAN Name
Add
Remove
VLAN No VLAN Name
1 default
10 VLAN10
20 VLAN20
21 VLAN21
1002 fddi-default
Equivalent IOS Commands
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#


Switch0
Physical
Config
CLI
IOS Command Line Interface
Switch>show vlan brief
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa3/1, Fa4/1, Fa5/1
10 VLAN10 active
11 VLAN11 active
20 VLAN20 active Fa0/1, Fa1/1, Fa2/1
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch>show interfaces trunk
Switch>
Copy
Paste

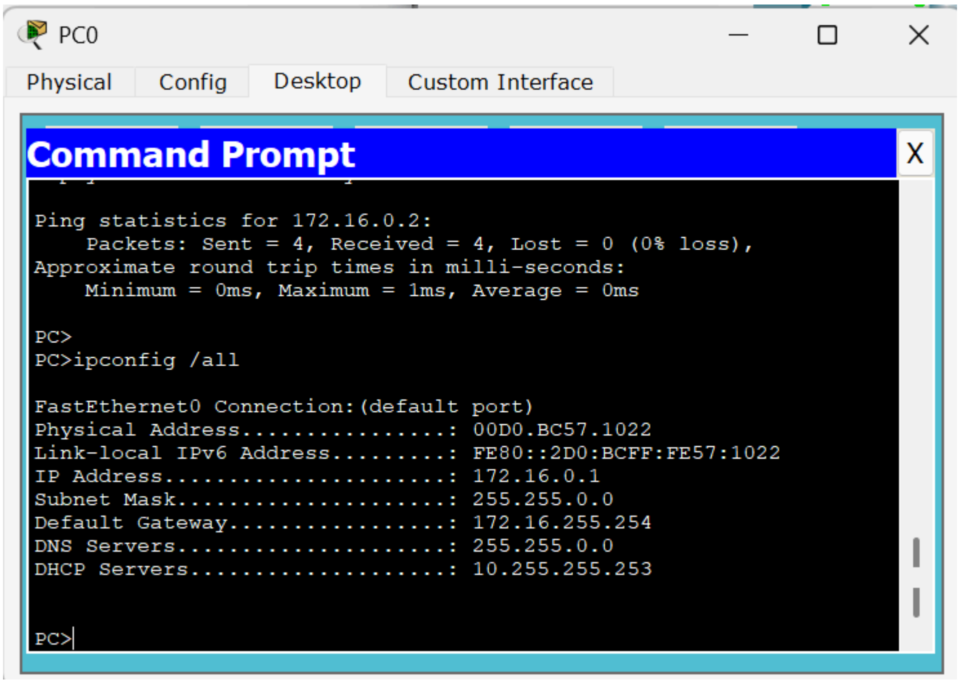

Switch1
Physical
Config
CLI
IOS Command Line Interface
21 VLAN21 active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch>show vlan brief
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa3/1, Fa4/1, Fa5/1
10 VLAN10 active Fa0/1, Fa1/1, Fa2/1
20 VLAN20 active
21 VLAN21 active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch>
Copy
Paste

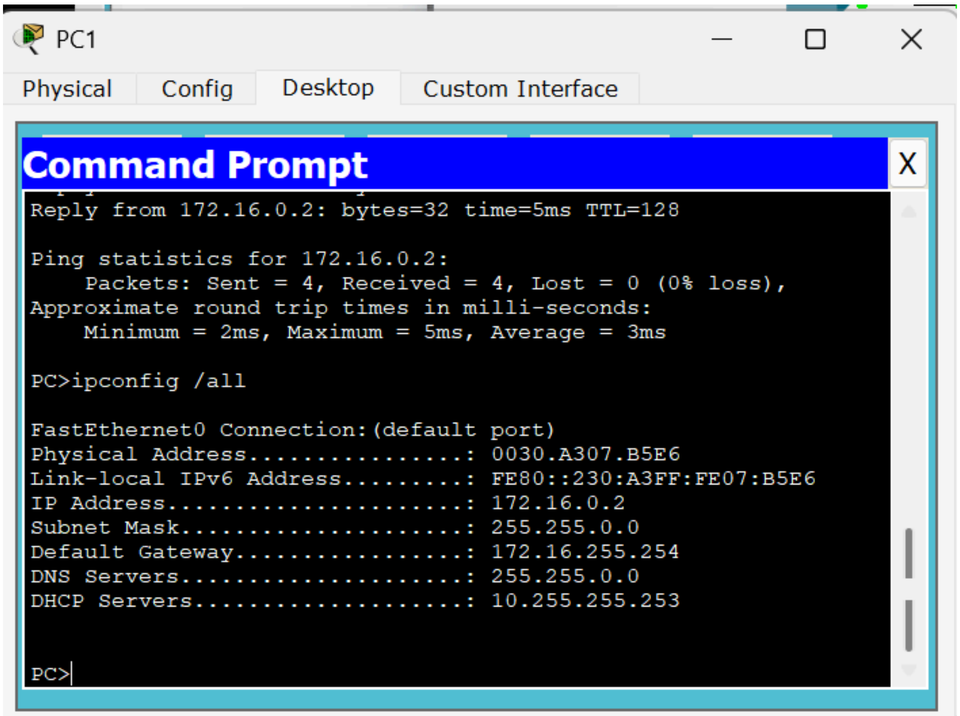
configuration du routeur :

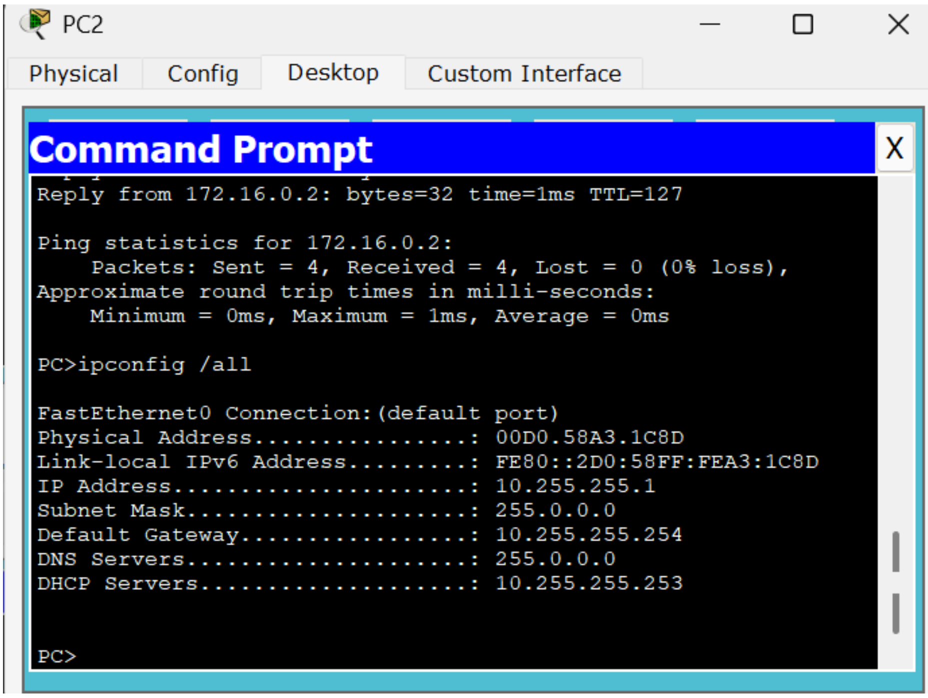




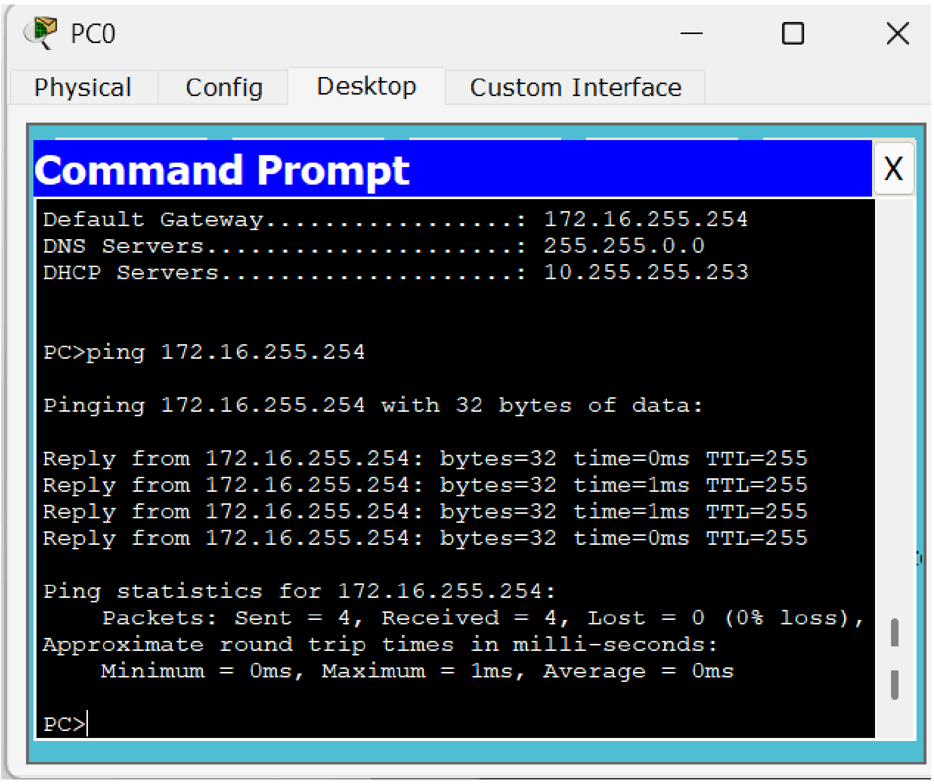
Test du DHCP sur les clients :

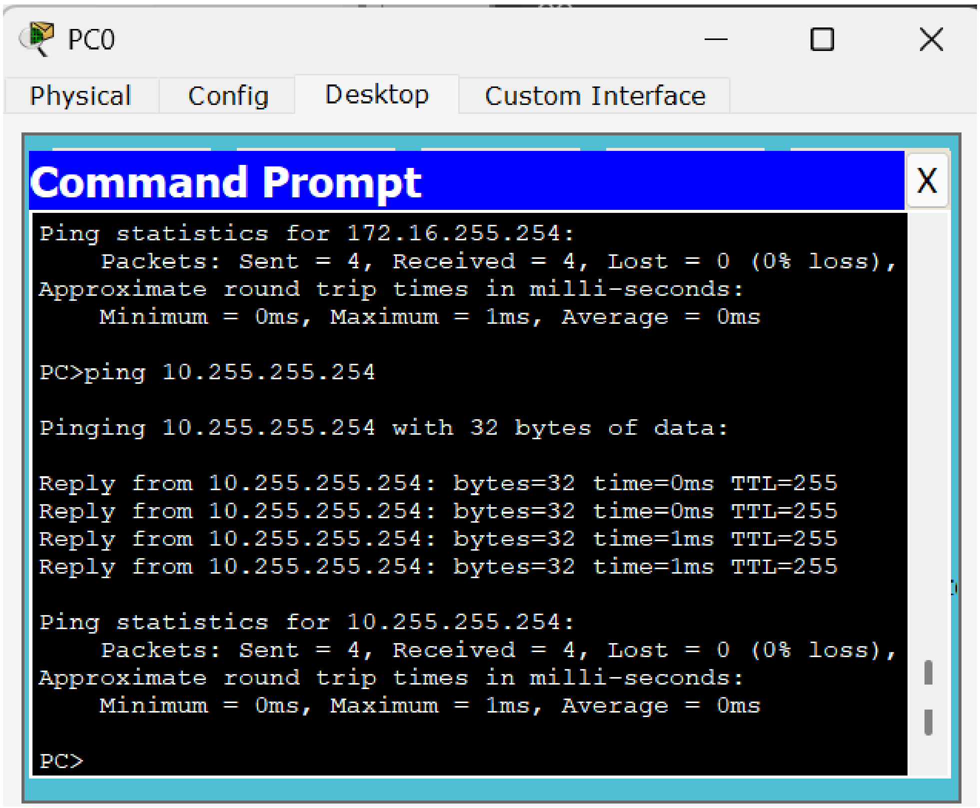


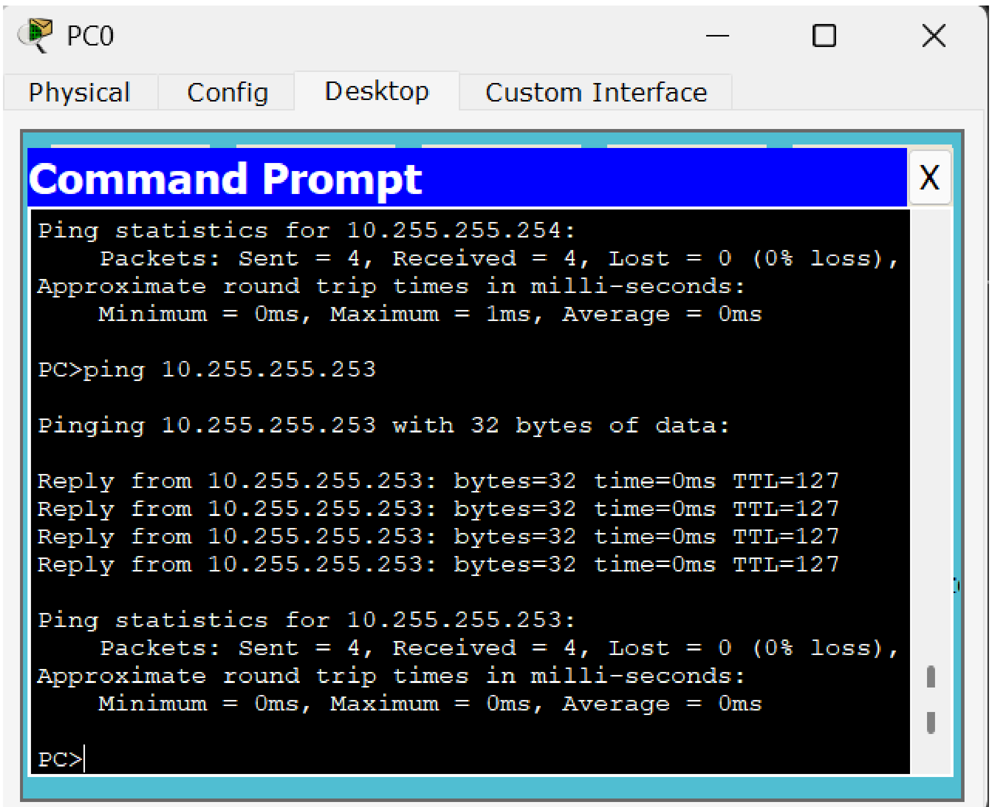


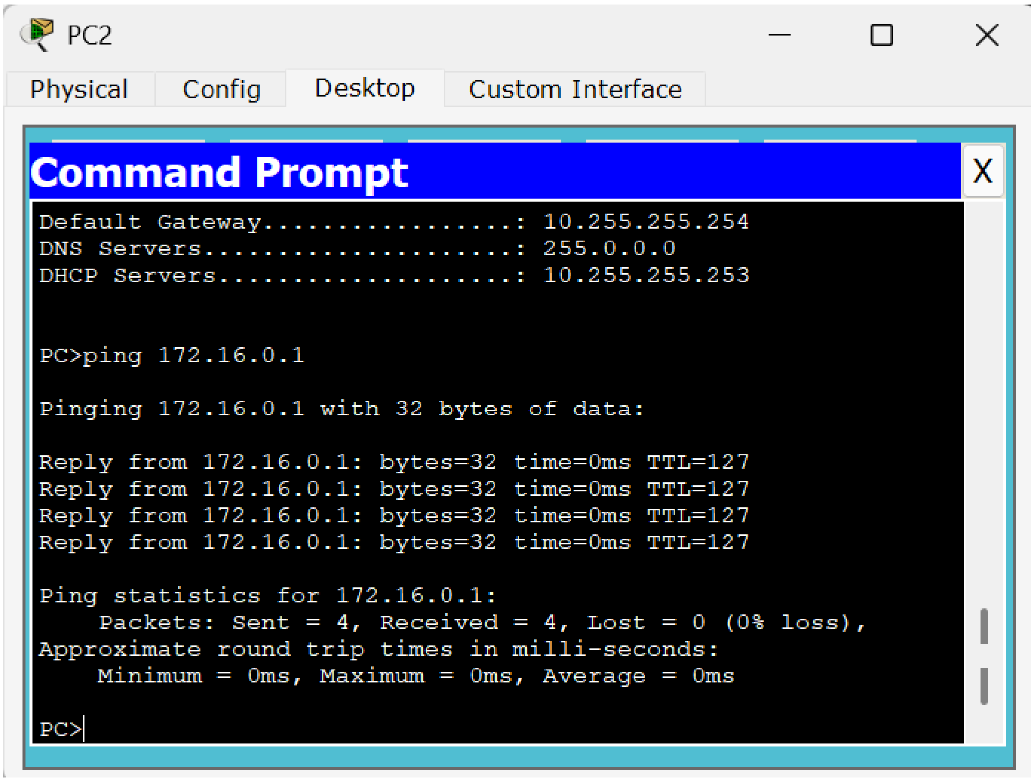


Test de pings :




PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
X
Ping statistics for 172.16.255.254:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>ping 10.255.255.254
Pinging 10.255.255.254 with 32 bytes of data:
Reply from 10.255.255.254: bytes=32 time=0ms TTL=255
Reply from 10.255.255.254: bytes=32 time=0ms TTL=255
Reply from 10.255.255.254: bytes=32 time=1ms TTL=255
Reply from 10.255.255.254: bytes=32 time=1ms TTL=255
Ping statistics for 10.255.255.254:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
X
Ping statistics for 10.255.255.254:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>ping 10.255.255.253
Pinging 10.255.255.253 with 32 bytes of data:
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Ping statistics for 10.255.255.253:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>



Tous les tests ont permis de vérifier que la configuration marche, le ping vers la passerelle de chaque VLAN répond correctement, ce qui confirme que les interfaces des routeurs sont bien joignables. Les pings entre hôtes de Vlan différents fonctionnent, ce qui prouve que le routage inter vlan marche. Le ping vers le serveur DHCP depuis le vlan20 marche également ce qui montre que l'ip helper transmet bien les requêtes entre les deux réseaux.

Conclusion :

Ce TP visait à séparer le réseau en deux VLAN , VLAN 20 pour les client 0 et 1 et VLAN 10 pour le client 3 et le DHCP, et à assurer l'attribution automatique des adresses IP.
Les ports des switchs ont été affectés aux VLAN correspondants, puis le routeur a été configuré avec un passerelle pour chaque sous réseau afin de permettre le routage entre eux.Comme le serveur DHCP se trouvait dans un autre réseau l'agent relais DHCP (ip helper-address) à été mis en place pour transmettre les requêtes des clients du VLAN 20 vers le serveur. Les tests ont confirmé que les PC reçoivent une adresse IP correcte et que la communication entre les deux réseaux fonctionne